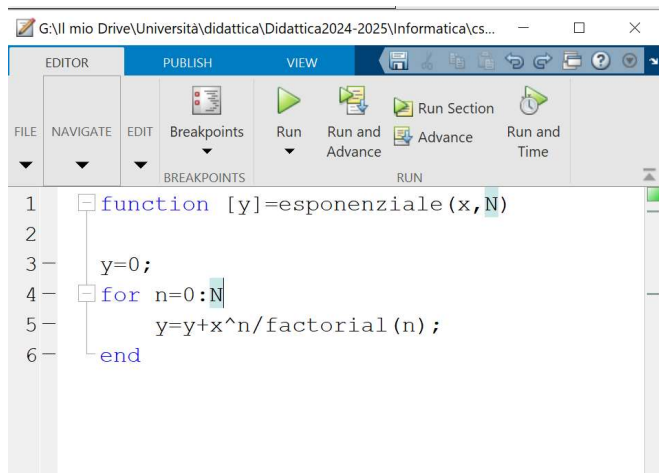


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for m=0:N
 termini
end

$$e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots + \frac{x^N}{N!} = \sum_{m=0}^N \frac{x^m}{m!}$$

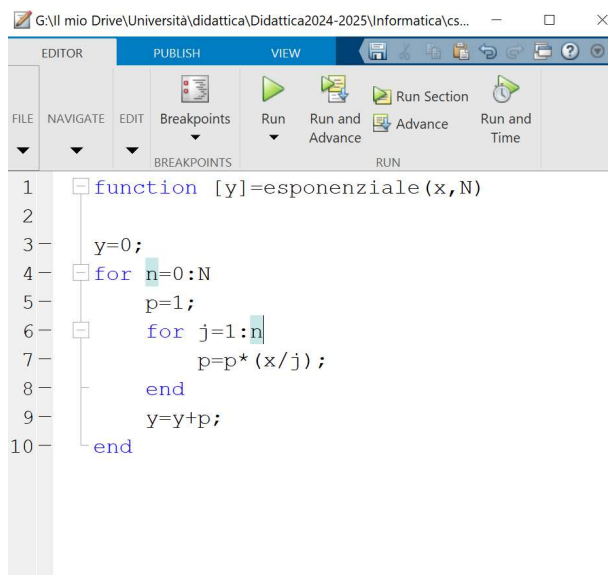


```
1 function [y]=esponenziale(x,N)
2
3     y=0;
4     for n=0:N
5         y=y+x^n/factorial(n);
6     end
```

per x ed n grandi $\Rightarrow x^n$ overflow

$n! = n(n-1)(n-2)\dots 1$ overflow

$$\frac{x^n}{n!} = \frac{x \cdot x \cdot x \dots x}{n \cdot (n-1) \cdot (n-2) \dots 1} = \left(\frac{x}{n}\right) \cdot \left(\frac{x}{n-1}\right) \cdot \left(\frac{x}{n-2}\right) \dots \left(\frac{x}{1}\right)$$



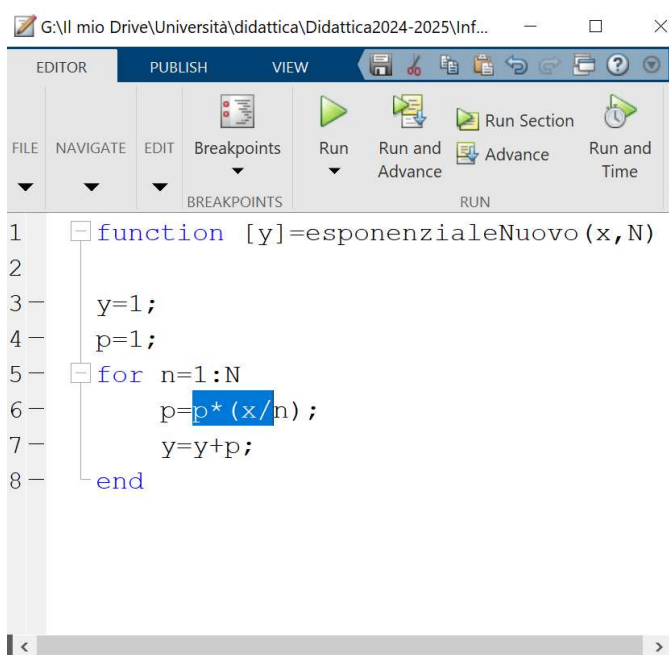
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BREAKPOINTS RUN

```
1 function [y]=esponenziale(x,N)
2
3     y=0;
4     for n=0:N
5         p=1;
6         for j=1:n
7             p=p*(x/j);
8         end
9         y=y+p;
10    end
```



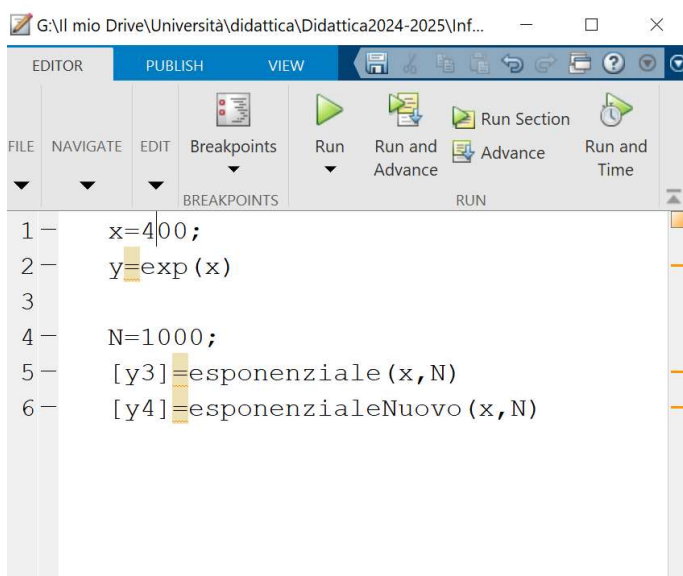
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BREAKPOINTS RUN

```
1 function [y]=esponenzialeNuovo(x,N)
2
3     y=1;
4     p=1;
5     for n=1:N
6         p=p*(x/n);
7         y=y+p;
8     end
```



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BREAKPOINTS RUN

```
1 x=400;
2 y=exp(x)
3
4 N=1000;
5 [y3]=esponenziale(x,N)
6 [y4]=esponenzialeNuovo(x,N)
```

```
Command Window
New to MATLAB? See resources for Getting Started.
5.221469689764124e+173

>> esercizio

y =

    5.221469689764144e+173

y3 =

    5.221469689764138e+173

y4 =

    5.221469689764138e+173
```